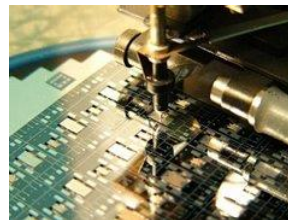




AlGaN/GaN-Based Transistor Development

Ohmic Contact and Schottky Gate Contact Characterization

Do Lai Tin Dat
dotindat@gmail.com
Supervisor: Dr. Péter Lajos Neumann



MOTIVATION

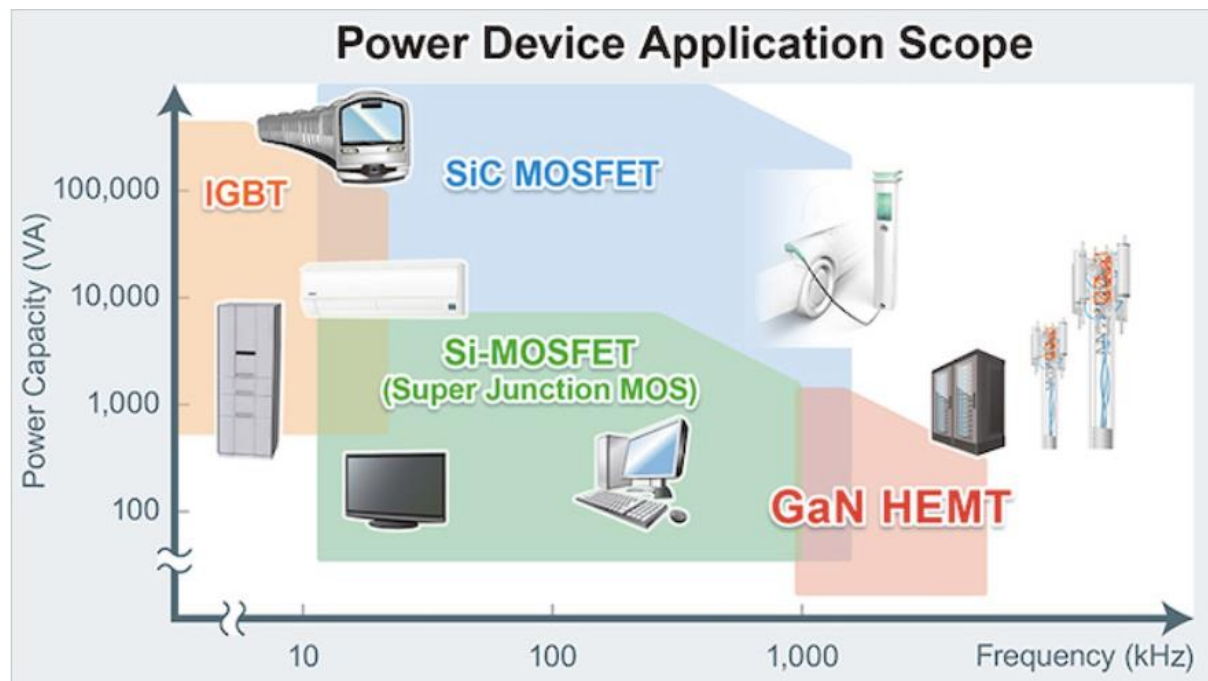
Material	Si	GaAs	4H-SiC	GaN
E_G (eV)	1.12	1.42	3.23	3.4
ϵ_r	11.7	12.9	9.66	8.9
μ (cm ² /V s)	1440	9400	950	1400
E_{crit} (MV/cm)	0.3	0.4	2.5	3.3
v_s ($\times 10^7$ cm/s)	1	0.9	2	2.4
κ_{th} (W/cm K)	1.3	0.55	3.7	2.5

Source: [1]

High v_s , and wide E_g make GaN-based devices potential for high-frequency, high-power applications.

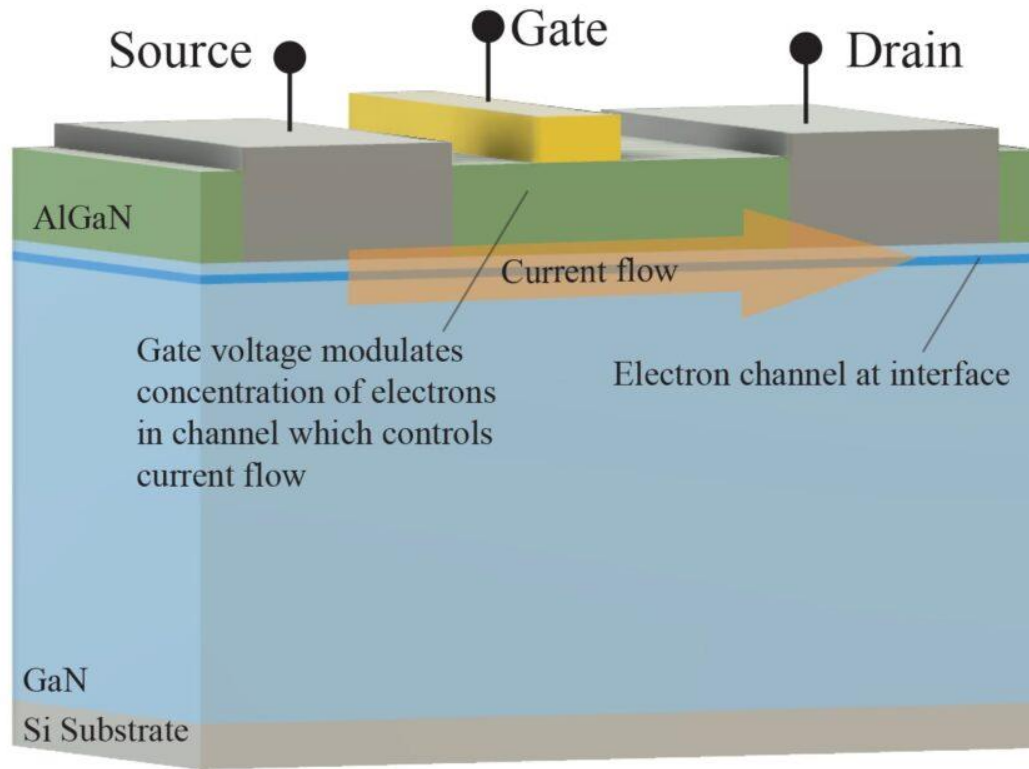
Advantage:

- Can operate in extreme environment (high T, radiation).
- Operate under high voltage.
- High saturation velocity.



Source: allaboutcircuits.com

DEVICE STRUCTURE

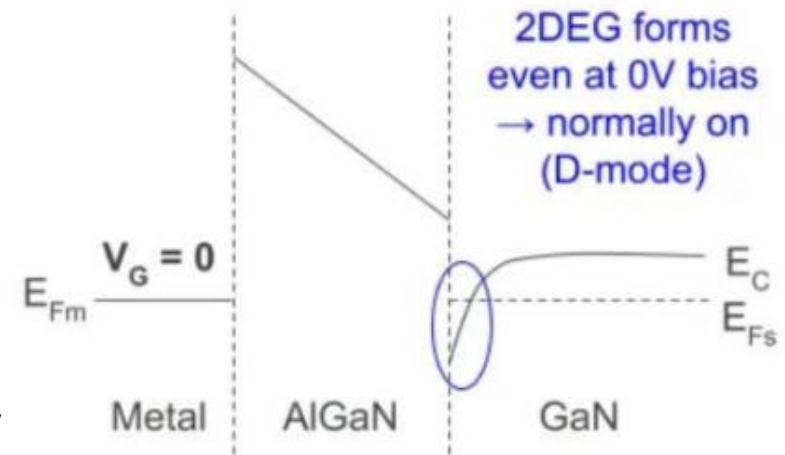


Source: semiengineering.com

A 2-dimension electron gas (2DEG) forms inherently at the AlGaN/GaN interface, due to

- Spontaneous polarization in each layer.
- Piezoelectric polarization from lattice mismatch.

AlGaN (25nm) and GaN (1000nm) layers are unintentionally doped, eliminating impurity scattering.

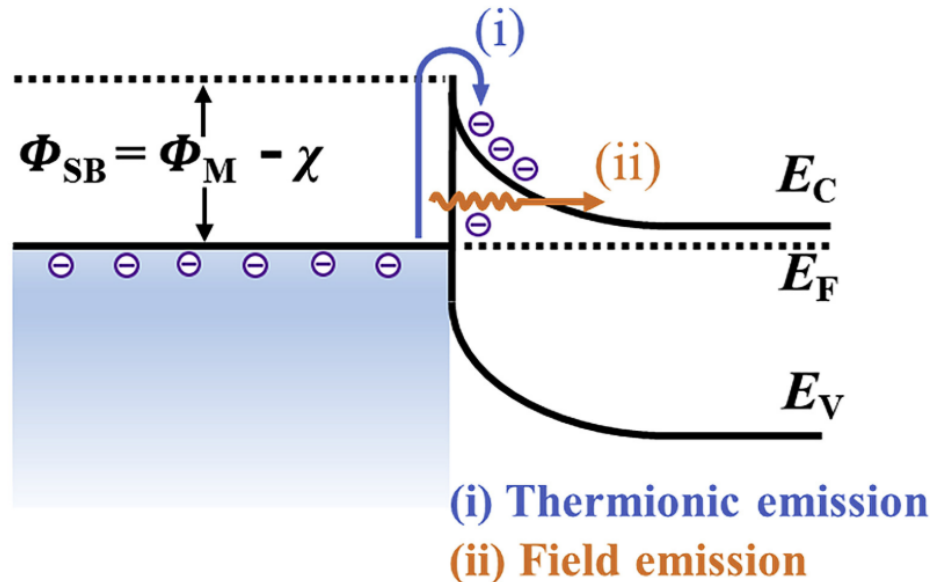


Source: Danielle-Gruber Blog

OHMIC CONTACT

The Ohmic contact is **a metal/semiconductor contact** that allows the current to flow freely in both directions, i.e., it is a non-rectifying contact.

In ideal case, the Schottky barrier height (SBH) is only dependent on material properties.



Source: [Zheng et al.](#)

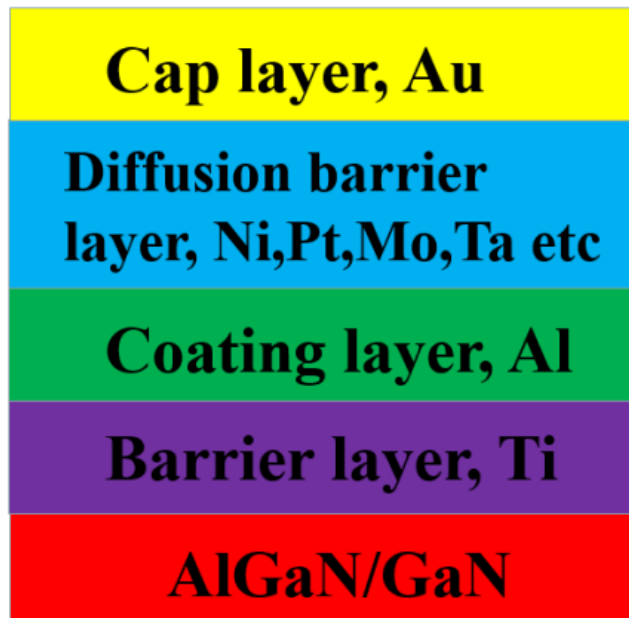
Approaches:

- Reduce Schottky barrier height by choosing metal with low work function.
- Reduce barrier thickness by increase doping concentration.

METALLIZATION SCHEME

Ti/Al/X/Au schemes dominate due to:

- Low work functions of Al and Ti.
- TiN and N-vacancies beneath contact.
- Ti prevents Al diffusion.
- High conductivity of Al.
- Oxidation-resistant and wire-bondable of Au.



Source: [2]

Metal Work Function ϕ_m	
Metal	[eV]
Ta	4.25
Al	4.28
Ti	4.33
Mo	4.6
Au	5.1
Pd	5.12
Ni	5.15
Ir	5.25
Pt	5.65

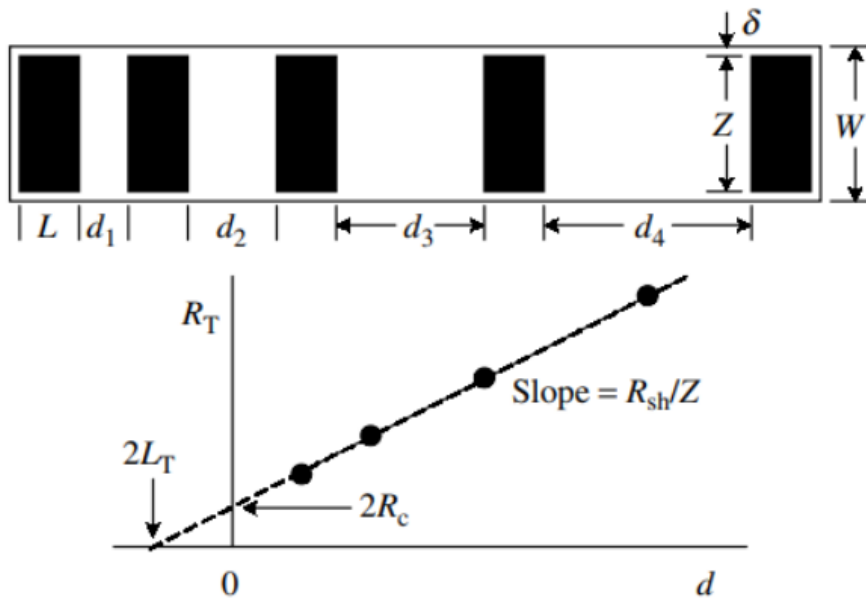
Source: [3]

In this work, **Ti/Al/Ti/Au (10nm/100nm/45nm/55nm)** was deposited using e-beam evaporation followed by RTA at 820°C for 90s in Nitrogen.

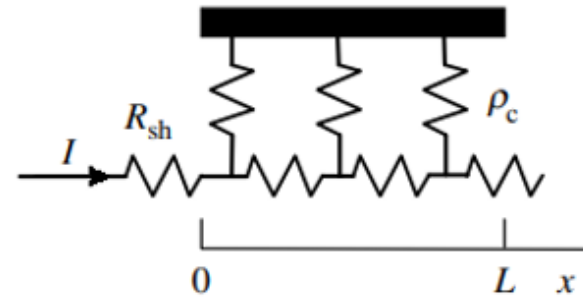
TRANSMISSION LENGTH MODEL (TLM)

Contact resistance R_c (Ω) and geometry-independent **specific contact resistance** ρ_c ($\Omega \cdot \text{cm}^2$) are key parameters used to evaluate ohmic contact quality.

TLM method is commonly used to determine these values.



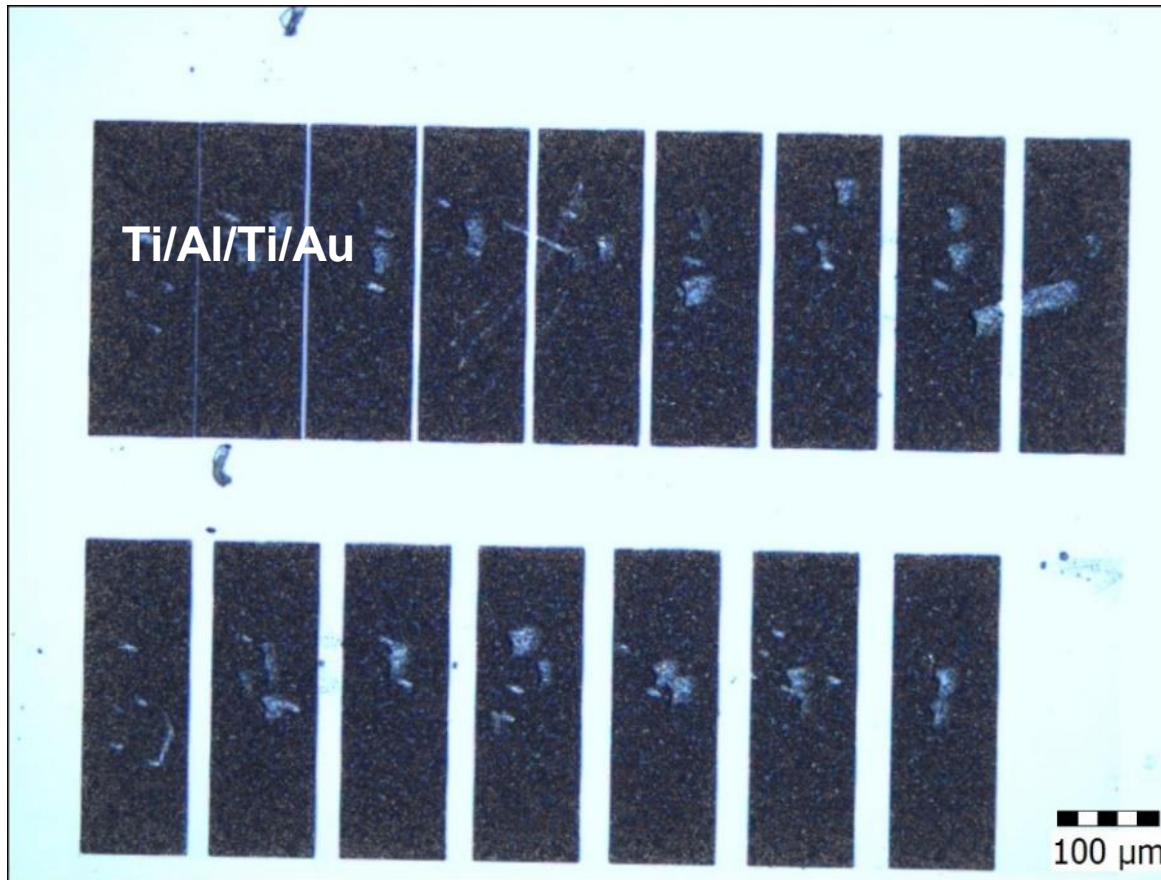
Source: [4]



$$R_T = \frac{R_{sh}d}{Z} + 2R_c \approx \frac{R_{sh}}{Z}(d + 2L_T)$$

TLM requires $Z = W$ condition, so mesa isolation is needed.

MEASUREMENT SETUP



Sample 1 (Au)

14 gaps with increasing spacings from 2 μm to 34 μm.

Pad width $Z = 300 \mu\text{m}$.
Pad length $L = 100 \mu\text{m}$.

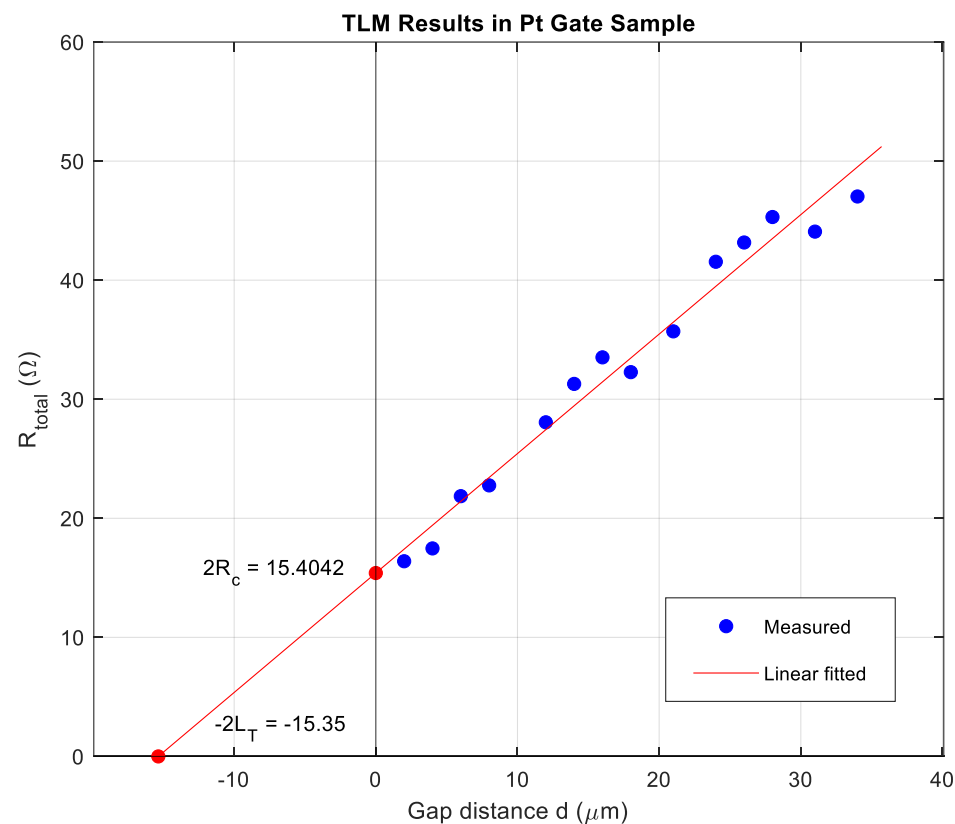
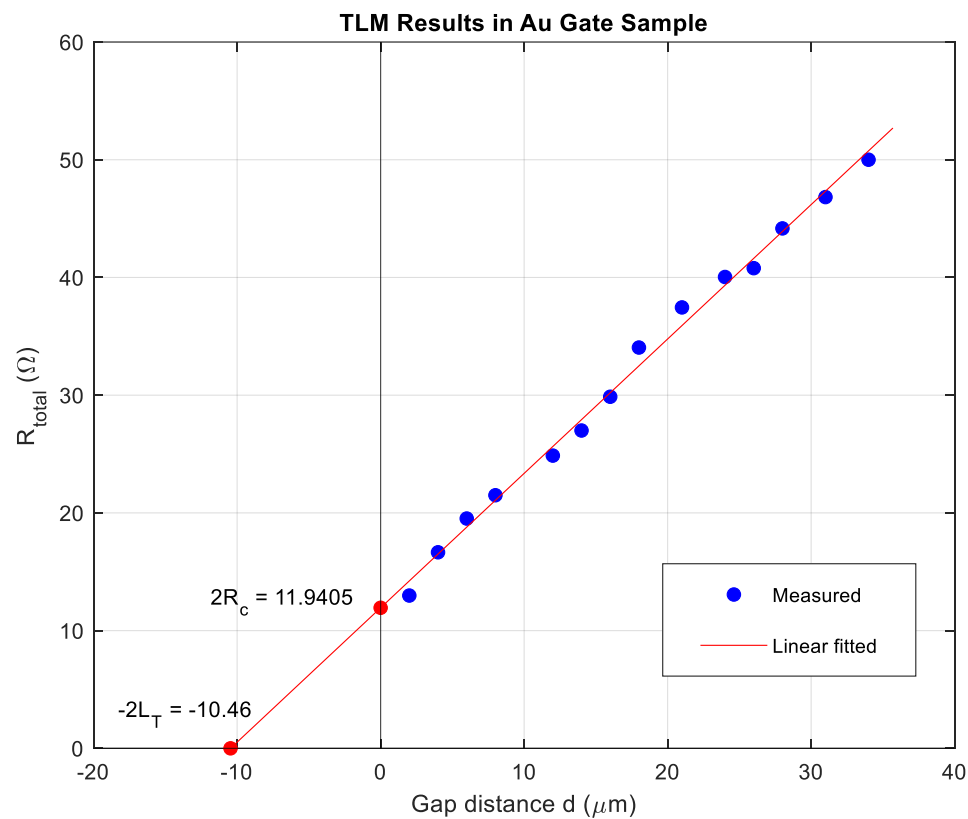
Mesa isolation $Z = W$, is verified.

Keithley 2450 setup:

- Source Current – Measure Voltage, validating linear V-I behavior.
- Sweep range: $-500 \mu\text{A}$ and $500 \mu\text{A}$; step: $20 \mu\text{A}$.

MATLAB was used for following mathematical steps.

TLM RESULTS



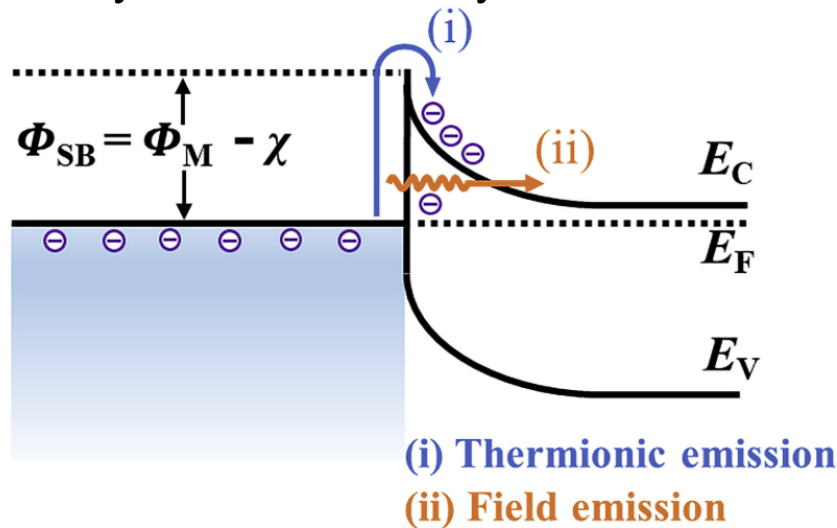
	Sample 1 (Au gate)	Sample 2 (Pt gate)
Contact resistance R_c (Ω)	5.97	7.70
Specific contact resistance ρ_c ($\Omega \cdot cm^2$)	9.39×10^{-5}	1.7×10^{-4}
Sheet resistance R_{sh} (Ω/sq)	342	300

Datasheet:
444 Ω/sq

SCHOTTKY GATE CONTACT

The Schottky gate contact is **a metal/semiconductor (M/S) contact** that allows the current to flow in only a direction, i.e., it is a rectifying contact.

Schottky contact naturally forms at M/S interface.



Source: [Zheng et al.](#)

Metal Work Function Φ_m	
Metal	[eV]
Ta	4.25
Al	4.28
Ti	4.33
Mo	4.6
Au	5.1
Pd	5.12
Ni	5.15
Ir	5.25
Pt	5.65

Source: [3]

Due to non-ideal M/S interface (such as interfacial layer, interface state)
-> The SBH cannot reach the ideal value.

Approaches:

- Increase Schottky barrier height by choosing metal with high work function.
- Increase the surface quality through surface treatments.

THERMIONIC EMISSION MODEL

Many gate parameters can be extracted from forward **I-V characteristics**.

Schottky gate contact can be modeled as a Schottky diode, which is described by:

$$I = I_S \cdot \left[\exp \left(\frac{q(V - IR_S)}{nkT} \right) - 1 \right] \quad (1) \quad \text{where } \Phi_B \text{ is Schottky barrier height (eV).}$$

n is ideality factor.

$$I_S = A_{eff} \cdot A^* \cdot T^2 \cdot \exp \left(\frac{-q\Phi_b}{kT} \right) \quad (2)$$

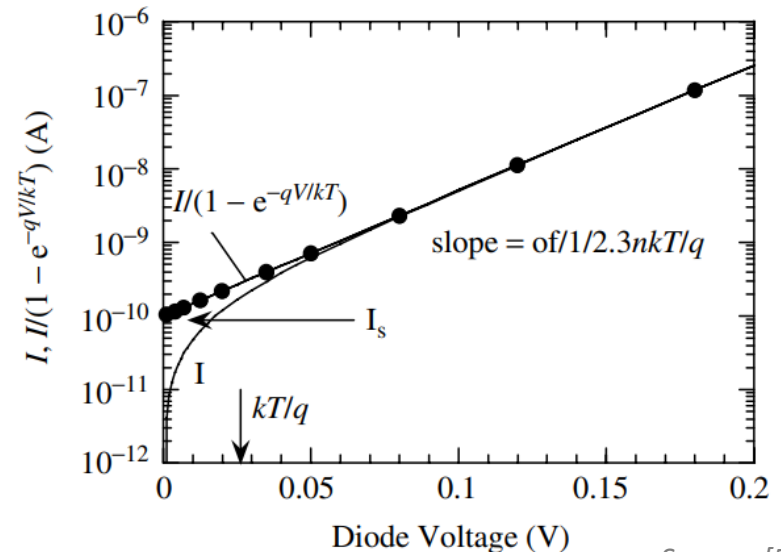
A_{eff} is effective area (cm^2).
 $A^* = 26.4 \frac{A}{cm^2 K^2}$ is Richardson constant.
 R_S is series resistance.

If $IR_S \ll V$ and $V \gg \frac{kT}{q}$, then the Eq. (1) becomes:

$$\ln(I) = \ln(I_S) + \frac{q}{nkT} V$$

This is purely thermionic emission (TE) method

In this work, A_{eff} is considered as constant.



Source: [5]

CHEUNG'S METHOD

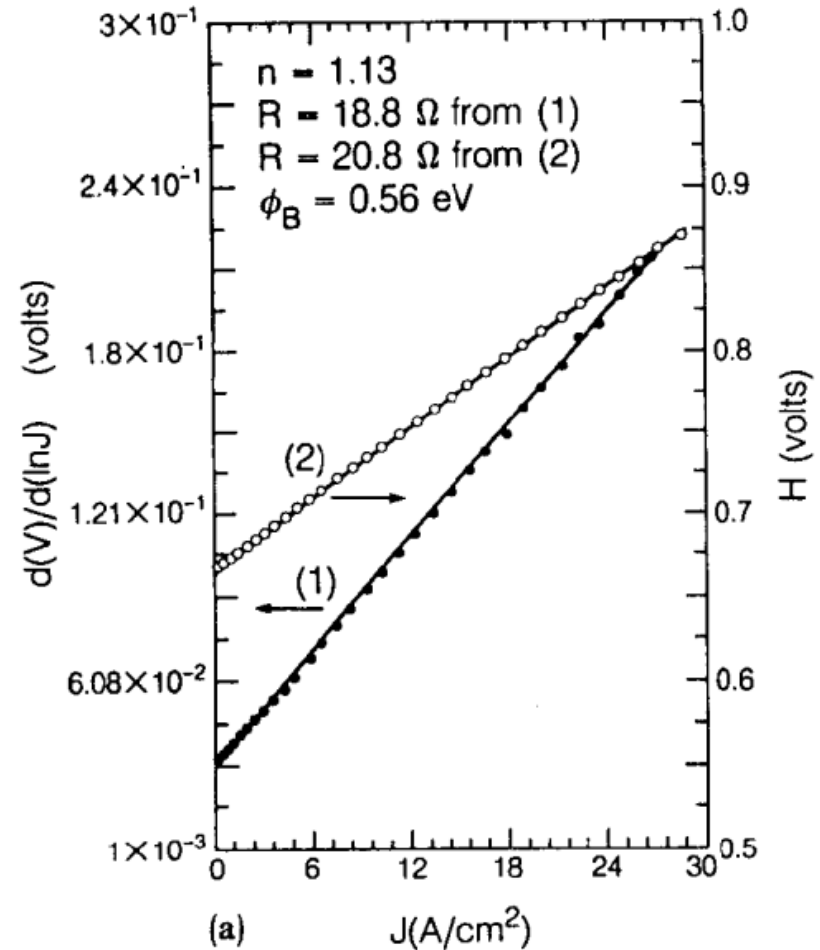
In case of high forward bias

If $V \gg \frac{kT}{q}$, Eq. (1) and (2) become

$$\frac{dV}{d(\ln I)} = R_s I + \frac{kT}{q} n$$

$$H(I) = R_s I + n\Phi_B$$

$$\text{where } H(I) = V - \frac{nkT}{q} \ln \left(\frac{I}{A_{eff} A^* T^2} \right).$$

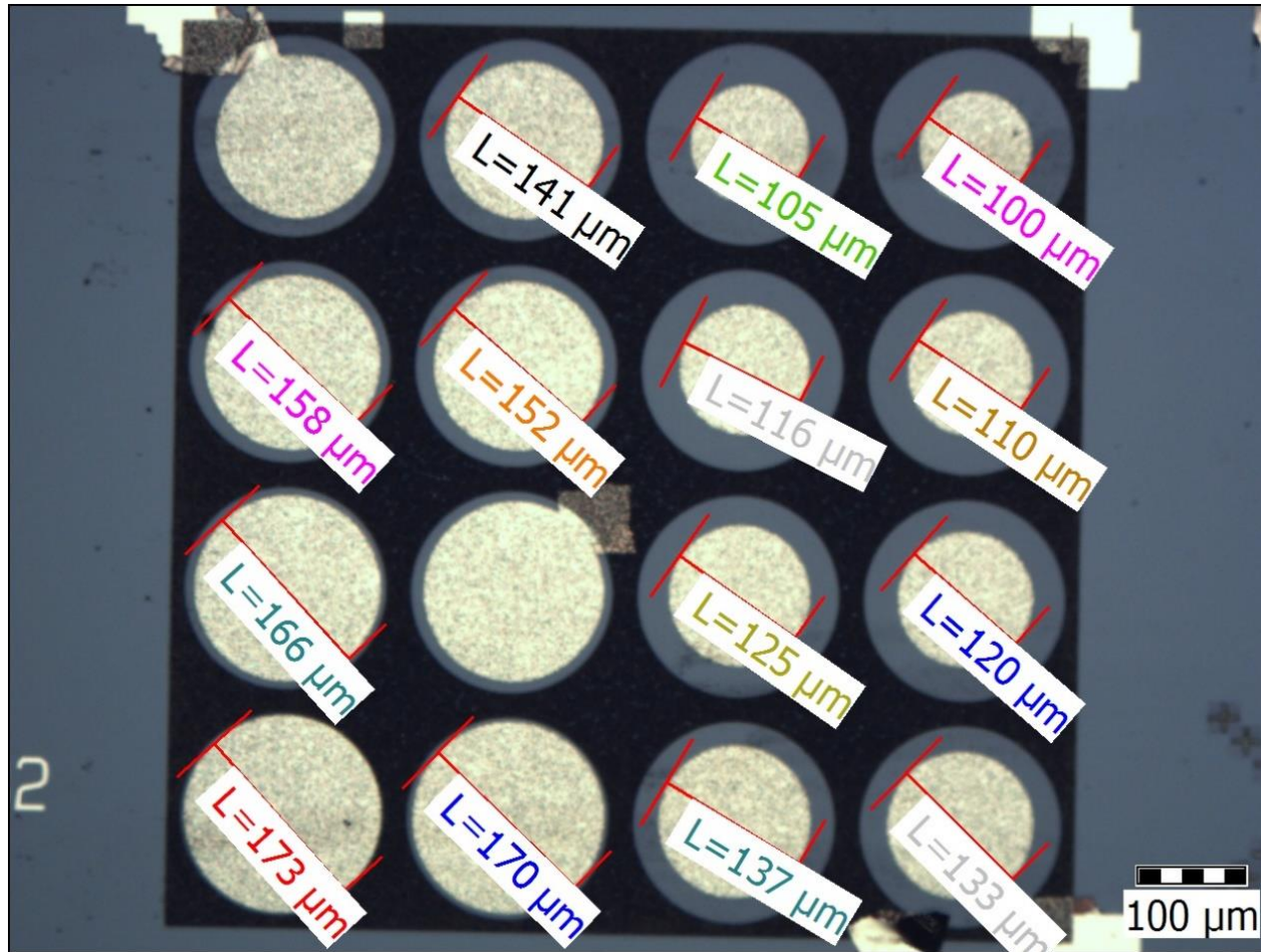


Source: [5]

In this work, A_{eff} is considered as constant.

MEASUREMENT SETUP

The CTLM structure consists of a circular inner Schottky gate contact surrounded by an outer ohmic circle, forming a lateral circular Schottky diode.



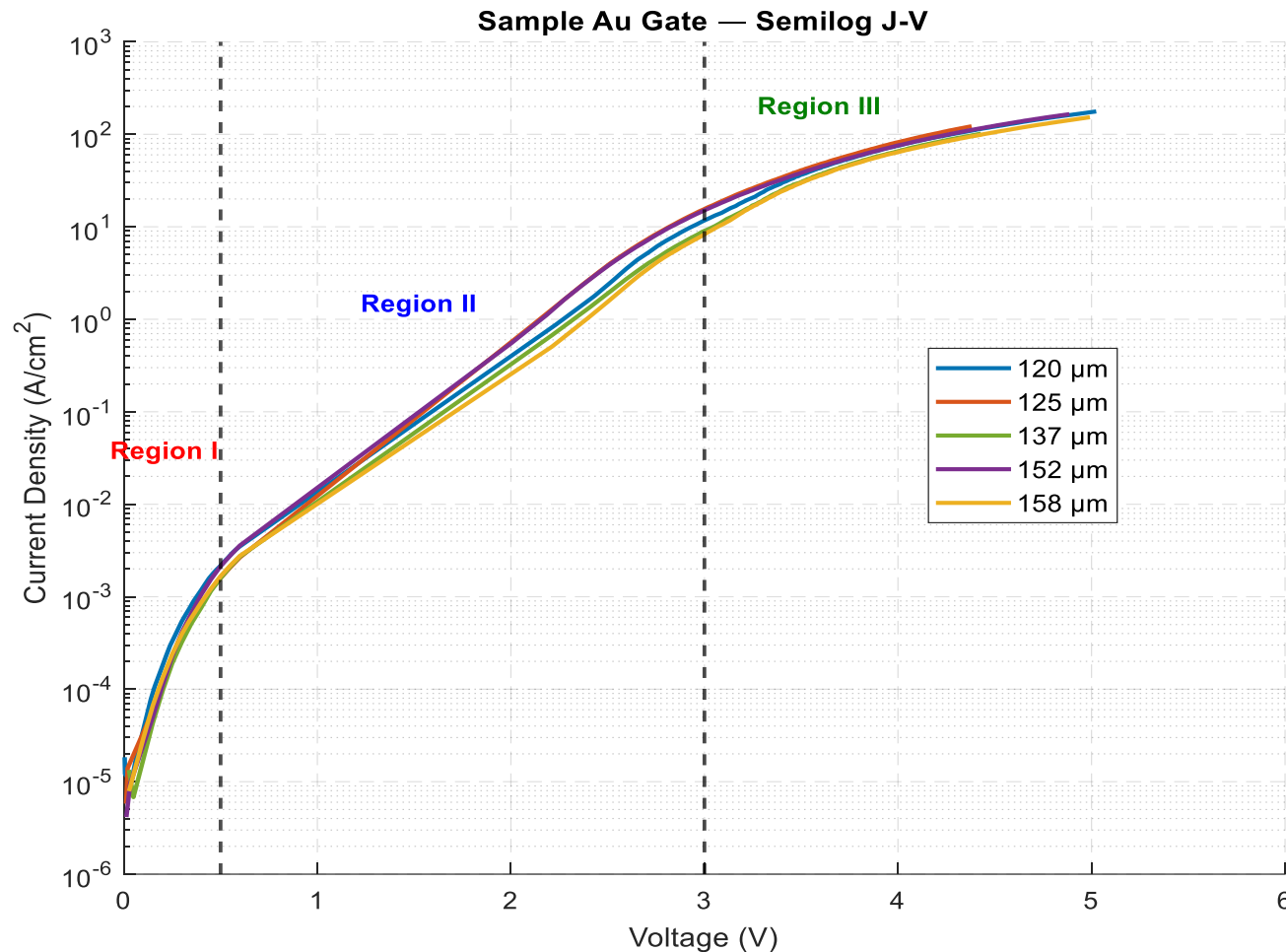
Sample 1 (Au)

I-V characteristics were achieved by Keithley 2450.

In this work, two samples – Au (150nm) and Pt (100nm) were deposited by e-beam evaporation – are investigated.

In sample Au, RIE was applied as a surface treatment.

SAMPLE AU: J-V CHARACTERISTICS



Region 1: shunt resistance dominates, $I = V/R_{sh}$ [6].

Region 2: effect of series resistance is negligible. So, purely TE method is applied here.

Region 3: series resistance reduces the increase in current. So, Chueng's method is applied here.

PURELY TE METHOD'S RESULT

Diameter (μm)	Ideality factor n	SBH Φ_B (eV)
120	11.88	0.58
125	11.69	0.58
137	11.56	0.58
152	12.58	0.57
158	12.39	0.58

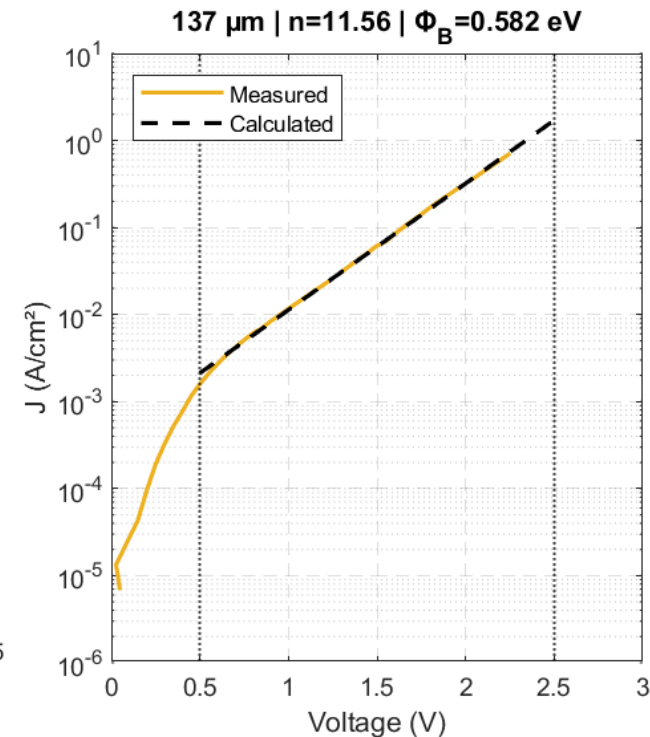
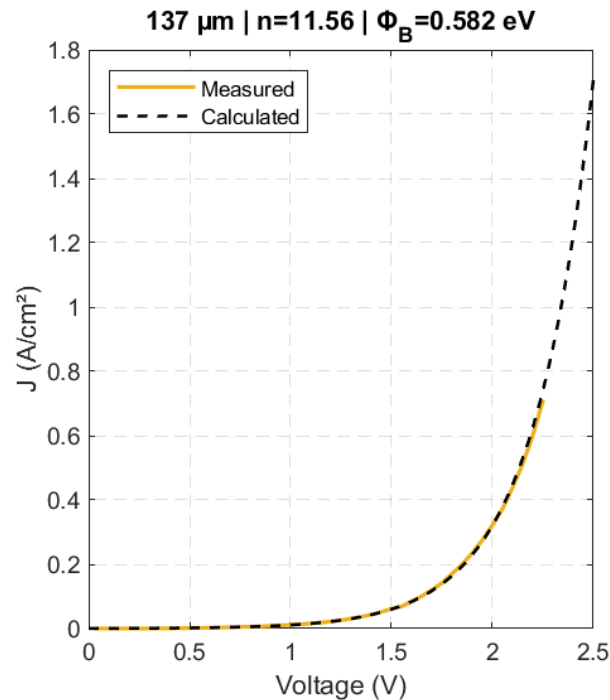
The ideality factor values (~ 12) are too high.

SBHs (~ 0.58) are too low.

Expected values:

$$n \approx 1 - 3$$

$$\Phi_B \approx 0.8 - 1.3 \text{ eV}$$



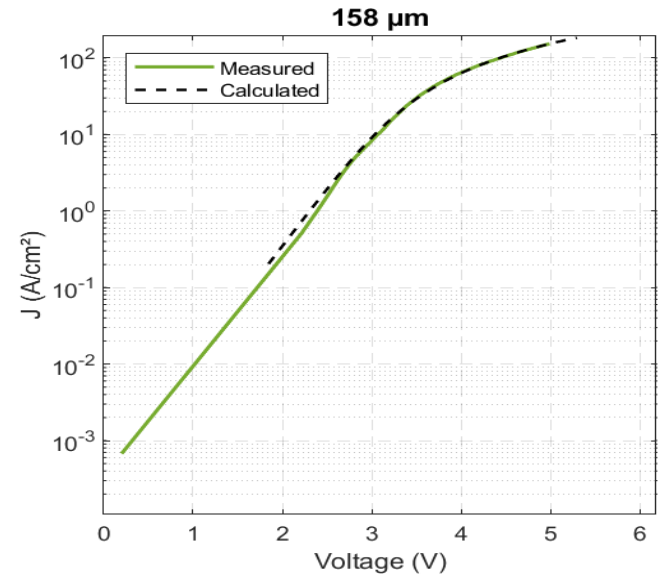
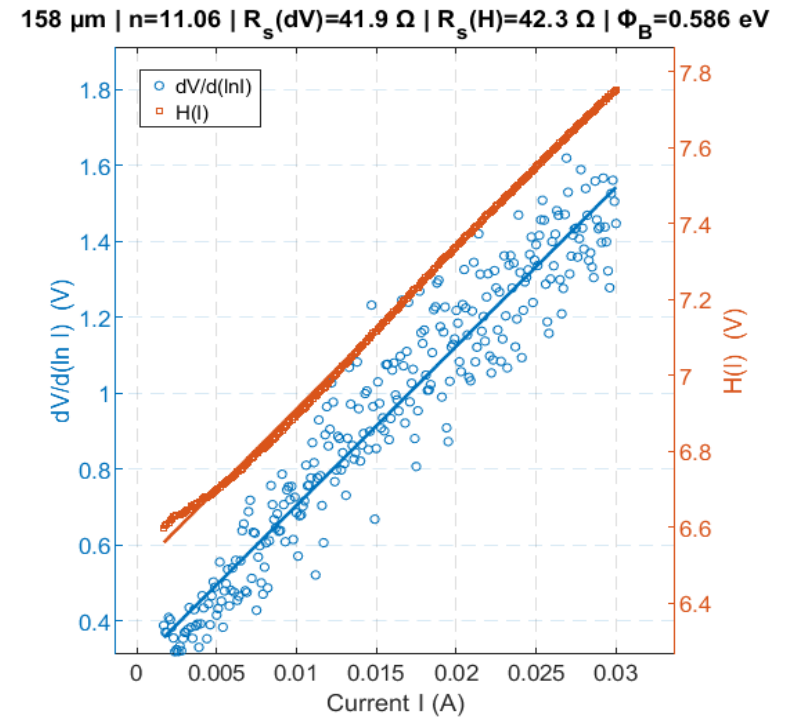
CHEUNG METHOD'S RESULT

Diameter (μm)	Ideality factor n	SBH Φ_B (eV)	Series resistance $R_s(\Omega)$ from $H(I)$ vs I plot	Series resistance $R_s(\Omega)$ from $dV/d(\ln I)$ vs I plot
120	11.11	0.58	67.21	67.00
125	14.18	0.51	47.73	47.85
137	9.20	0.64	60.96	61.43
152	14.10	0.52	36.97	37.03
158	11.06	0.59	42.28	41.90

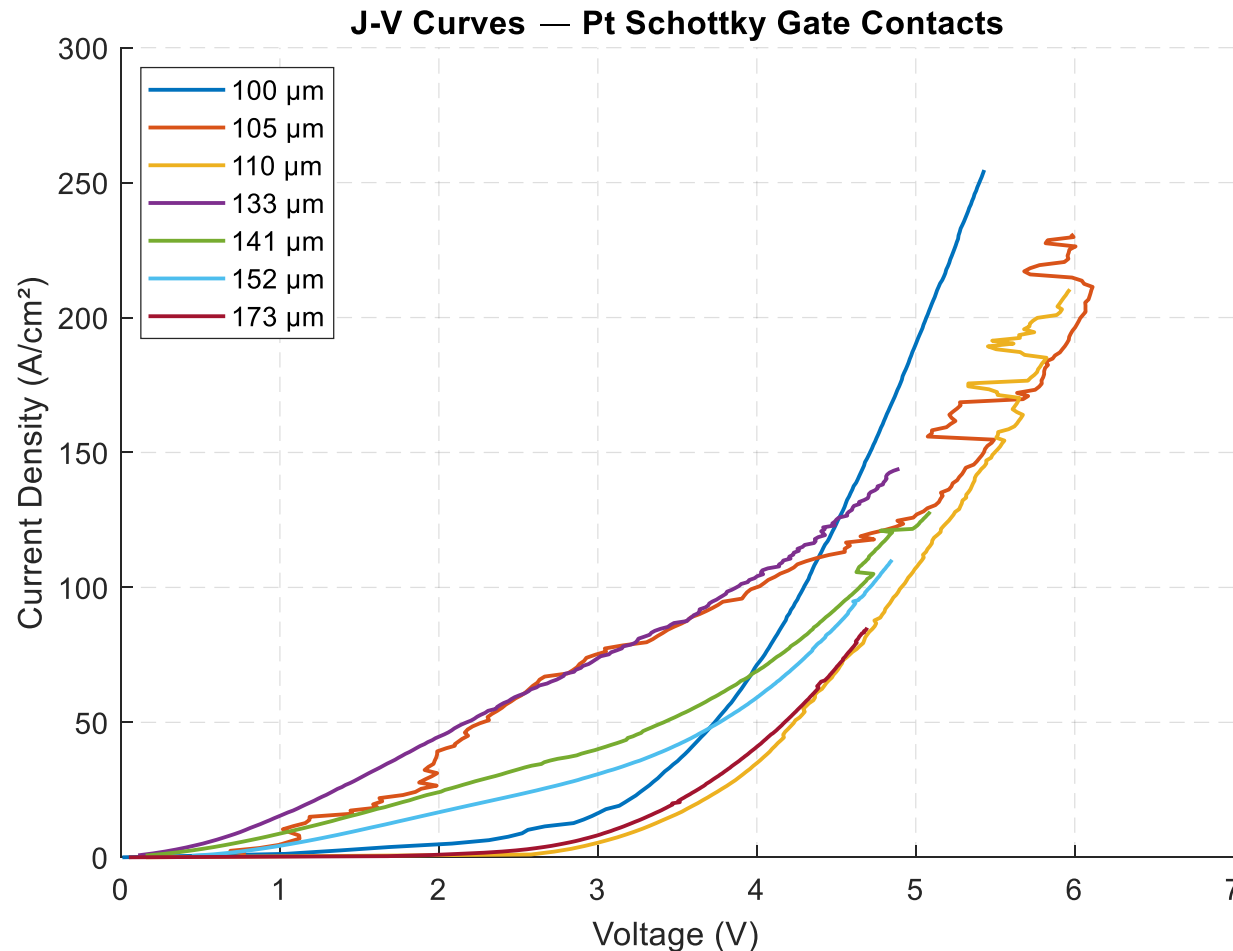
These results are consistent with previous.

This behavior can be due to impurity deposited metal, very non-ideal interface, or inappropriate voltage range for extraction methods.

However, further investigation is needed to determine this problem.



SAMPLE PT: J-V CHARACTERISTICS



Non-uniformity across Pt gates can be observed by J-V characteristics.

This non-uniformity can be attributed to the absence of RIE surface treatment.

This result highlights the importance of surface preparation in Schottky gate formation of AlGaIn/GaN structure.

NEXT STEPS

- Further investigation to determine the problem and achieve reliable Schottky gate quality.
- The effect of other device parameters, such as L_g, L_{gd}, L_{gs} , on device performance is investigated through three-terminal measurements, such as output characteristics and transfer characteristics.

Thank you for your attention

REFERENCES

- [1] Meneghini et al., “GaN-based power devices: Physics, reliability, and perspectives,” *Journal of Applied Physics*, vol. 130, no. 18, p. 181101, Nov. 2021, doi: 10.1063/5.0061354.
- [2] Greco et al., “Ohmic contacts to Gallium Nitride materials,” *Applied Surface Science*, vol. 383, pp. 324–345, Oct. 2016, doi: 10.1016/j.apsusc.2016.04.016.
- [3] Ofuonye et al., “Electrical and microstructural properties of thermally annealed Ni/Au and Ni/Pt/Au Schottky contacts on AlGaN/GaN heterostructures,” *Semicond. Sci. Technol.*, vol. 29, no. 9, p. 095005, Sep. 2014, doi: 10.1088/0268-1242/29/9/095005
- [4] Schroder, *Semiconductor Material and Device Characterization*, 1st ed. Wiley, 2005. doi: 10.1002/0471749095.
- [5] S. K. Cheung et N. W. Cheung, "Extraction of Schottky diode parameters from forward current-voltage characteristics," *Applied Physics Letters*, vol. 49, no. 2, pp. 85–87, Jul. 1986, doi: 10.1063/1.97359.
- [6] I. Jabbari et al., “Schottky barrier inhomogeneity in (Pd / Au) Al_{0.22} Ga_{0.78}N/GaN/SiC HEMT: Triple Gaussian distributions,” *Chinese Journal of Physics*, vol. 73, pp. 719–731, Oct. 2021, doi: 10.1016/j.cjph.2021.08.011.